

2(a)

award mp1 and mp8 anywhere in each section of the answer

5

cell B - mesophyll cells

mp1 photosynthesis / described ;

e.g. conversion of carbon dioxide to sugars using light energy

synthesis of complex organic compounds from inorganic compounds using light energy

adaptation – mp2 and one from mp3 to mp7

mp2 chloroplasts / chlorophyll / chloroplast pigments	to absorb light ;
any one from:	
mp3 large vacuole	to keep chloroplasts at the periphery ;
mp4 starch grains	as store of products of photosynthesis ;
mp5 large / moist, surface / cell wall	for evaporation of water for <u>transpiration</u> (stream) ; A provides water for photosynthesis in context of water supply transpiration
mp6 thin cell wall	gas exchange / diffusion of gases ;
mp7 isodiametric / (roughly) spherical shape	prevents close packing / gives large air spaces ;

cell C - sieve tube element

mp8 transport / movement of, (named) sugars / (named) assimilates / photosynthates / organic substances ;

A translocation

Question	Answer	Marks														
2(a)	<i>adaptation – any two from:</i> <table><tr><td>mp9 peripheral / described, cytoplasm A 'at the edge' / 'thin layer'</td><td>allowing maximum, volume / space, for transport of, phloem sap / assimilates ; A less / little, resistance to flow / AW</td></tr><tr><td>mp10 no nucleus / few organelles / little cytoplasm R 'no organelles'</td><td>allowing maximum, volume / space, for transport of, phloem sap / assimilates ; A less / little, resistance to flow / AW</td></tr><tr><td>mp11 elongate / elongated (cells, so end to end)</td><td>to make (sieve) tubes ;</td></tr><tr><td>mp12 plasmodesmata</td><td>(un)loading from / AW, companion cells / cell A ; I 'between cells A and C'</td></tr><tr><td>mp13 sieve pores</td><td>A flow between sieve tube elements ; AW</td></tr><tr><td>mp14 sieve plates</td><td>prevent, bursting / bulging ; A maintaining hydrostatic pressure</td></tr><tr><td colspan="2">mp15 AVP for C ; e.g. phloem proteins and defence against pathogens / sealing wounds e.g. cell walls with cellulose microfibrils that run around the cells and to prevent bursting</td></tr></table>	mp9 peripheral / described, cytoplasm A 'at the edge' / 'thin layer'	allowing maximum, volume / space, for transport of, phloem sap / assimilates ; A less / little, resistance to flow / AW	mp10 no nucleus / few organelles / little cytoplasm R 'no organelles'	allowing maximum, volume / space, for transport of, phloem sap / assimilates ; A less / little, resistance to flow / AW	mp11 elongate / elongated (cells, so end to end)	to make (sieve) tubes ;	mp12 plasmodesmata	(un)loading from / AW, companion cells / cell A ; I 'between cells A and C'	mp13 sieve pores	A flow between sieve tube elements ; AW	mp14 sieve plates	prevent, bursting / bulging ; A maintaining hydrostatic pressure	mp15 AVP for C ; e.g. phloem proteins and defence against pathogens / sealing wounds e.g. cell walls with cellulose microfibrils that run around the cells and to prevent bursting		
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2(b)	<i>mp1, 2, 4 and 6 needs idea of increases / more; this is not needed for mp3 and mp5</i> <i>any three from:</i> 1 increases surface area:volume <u>of cell</u> / increases surface area of (cell surface) <u>membrane</u> ; 2 <i>idea of</i> more space for / increased area for / more, proton pumps / carrier proteins in context of moving protons ; 3 pumping protons, from cytoplasm / into cell wall / into apoplast / AW ; A create / increase, proton / electrochemical / concentration, gradient 4 <i>idea of</i> more space for / increased area for / more, <u>cotransporter</u> proteins / carrier proteins in context ; 5 <u>cotransport</u> of sucrose / described, into companion cell / into transfer cell / into cell A / from mesophyll cell / from cell B ; A secondary active transport A <i>movement of sucrose <u>and</u> protons for cotransport</i> 6 more space for plasmodesmata (between cell A and cell C) ;	3